VIT Vidyalankar Institute of Technology
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Department of Computer Engineering

Semester	B.E. Semester VII – Computer Engineering
Subject	R Programming
Subject Professor In- charge	Prof. Sanjeev Dwivedi
Assisting Teachers	Prof. Sanjeev Dwivedi
Laboratory	M-411B

Student Name	Sneha Gadhari	
Roll Number	23102B0055	
Grade and Subject Teacher's Signature		

Experiment Number	01
Experiment Title	Air-Quality Data Cleaning Using R: Management of NA, NULL, and NaN Values

Resources / Apparatus Required	Hardware: Computer system	Software: Colab

Objective	An environmental monitoring agency collects hourly air-quality and weather readings from multiple monitoring stations. Due to sensor failures, temporary network interruptions, equipment maintenance, and data-entry errors, several pollutant and weather readings are unavailable or invalid. Before this data can be used for pollution analysis, it must be cleaned and validated. This practical uses the **Beijing Multi-Site Air Quality Dataset** (specifically the Aotizhongxin station, PRSA_Data_Aotizhongxin_20130301-20170228.csv), which contains 35,064 hourly observations across 18 variables including PM2.5, PM10, SO2, NO2, CO, O3, TEMP, PRES, DEWP, RAIN, wd, and WSPM. Some observations contain missing values represented as NA. The objective is to develop an R program that can: • Identify missing and undefined values (NA, NULL, NaN) • Differentiate between these three types of missing/undefined data • Summarize missing values across selected variables • Replace missing numerical values using suitable statistical methods • Replace missing categorical values appropriately • Handle errors gracefully without terminating the program • Export a cleaned dataset for further analysis
Description	Task: 1. Import and Inspect the Dataset 2. Understand NA, NULL, and NaN 3. Create a Missing-Value Summary Function 4. Identify Invalid Numerical Results 5. Handle Missing Numerical Values Using a Loop 6. Handle Missing Categorical Values 7. Implement Error Handling 8. Compare Missing Values Before and After Cleaning 9. Generate One Visualization 10. Export the Cleaned Dataset Github link: https://github.com/Sneha-Gadhari/R_Programming/tree/ad68ba9e54b154ff2944504e31a084fe8c234af7/Experiment-1

Output

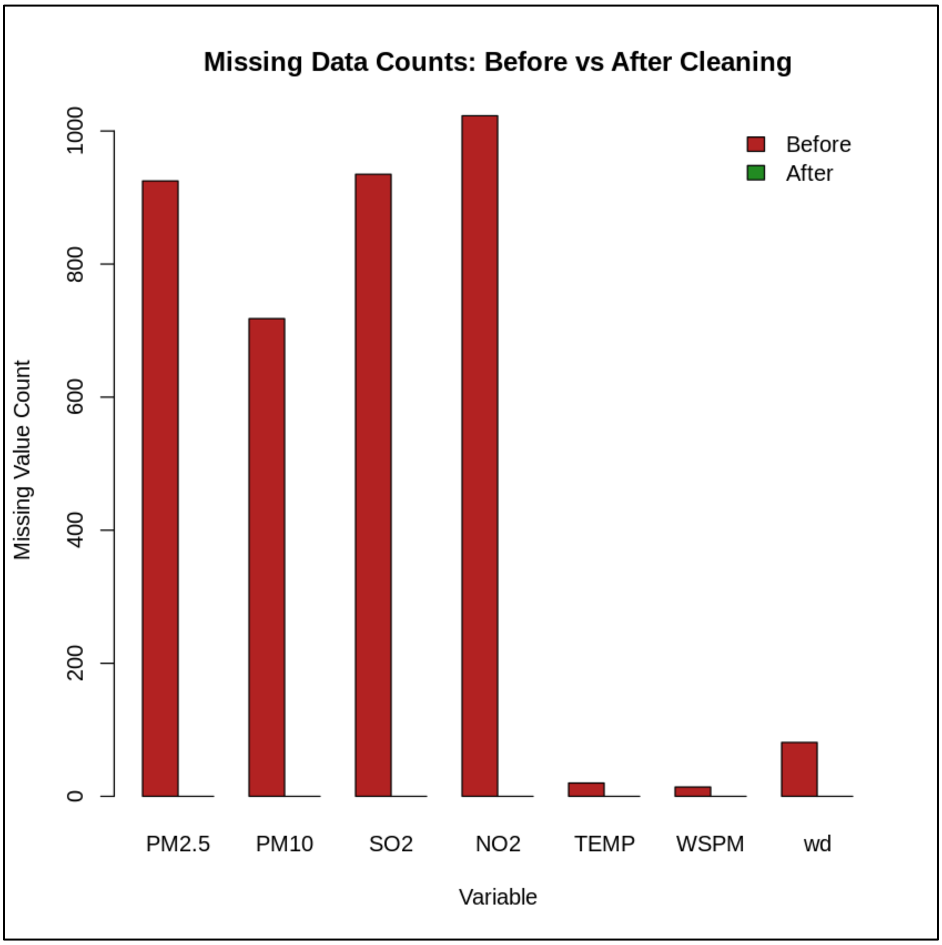
Task 3:

```
  Variable Total_Records Missing_Values Missing_Percentage
1    PM2.5         35064            925               2.64
2     PM10         35064            718               2.05
3      SO2         35064            935               2.67
4      NO2         35064           1023               2.92
5     TEMP         35064             20               0.06
6     WSPM         35064             14               0.04
7       wd         35064             81               0.23
```

Task 8:

```
  Variable Missing_Before Missing_After Values_Replaced
1    PM2.5            925             0             925
2     PM10            718             0             718
3      SO2            935             0             935
4      NO2           1023             0            1023
5     TEMP             20             0              20
6     WSPM             14             0              14
7       wd             81             0              81

Interpretation: Missing_After reads zero across every tracked variable,
confirming the imputation strategy (median for numeric fields, mode for wd) worked as intended.
```



Conclusion

This experiment implemented a complete missing-data handling pipeline in R on the Aotizhongxin station file of the Beijing air-quality dataset (35,064 observations across 18

	variables). The dataset was imported safely using tryCatch(), and the three special values NA, NULL, and NaN were demonstrated individually to clarify how R differentiates a missing observation, an absent object, and an undefined arithmetic result. A custom summarize_missing() function showed that missing values across PM2.5, PM10, SO2, NO2, TEMP, WSPM, and wd were all under 3% of total records — none breached the 20% warning threshold. The aqi_ratio variable (PM2.5/PM10) contained 934 NA values but no NaN or infinite results, so no additional replacements were needed there. A for-loop imputed missing values in the six numeric columns using their respective medians (e.g., 58 for PM2.5, 87 for PM10, 9 for SO2), while the categorical wd column's 81 missing entries were filled with its mode, "NE". The safe_clean_column() function correctly cleaned the valid numeric column SO2, while properly rejecting a categorical column (wd) and a nonexistent column (NoSuchColumn) with informative error messages instead of crashing. The before/after comparison table confirmed all 3,716 missing values across the seven tracked variables (925 + 718 + 935 + 1023 + 20 + 14 + 81) were fully resolved, visually verified by the bar chart, and the cleaned dataset was exported successfully. Overall, the practical reinforced core R skills in control structures, function design, and exception handling applied to a realistic data-cleaning problem.